Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

$\cos^{-1}\left[\cos\left(2\cot^{-1}(\sqrt{2}-1)\right]\right.$ is equal to

(A) $\sqrt{2}-1$

(B) $1-\sqrt{2}$

(C) $\frac{\pi}{4}$

(D) $\frac{3\pi}{4}$

Q.2

Solution set of inequation $\left(\cos^{-1}x\right)^2-\left(\sin^{-1}x\right)^2>0$ is

(A) $\left[0,\frac{1}{\sqrt{2}}\right)$

(B) $\left[-1,\frac{1}{\sqrt{2}}\right)$

(C) $(-1,\sqrt{2})$

(D) none of these

Q.3

The number of real solutions of $\cos^{-1}x+\cos^{-1}2x=-\pi$ is

(A) 0

(B) 1

(C) 2

(D) infinitely many

Q.4

If $\sin^{-1}x+\cot^{-1}\frac{1}{2}=\frac{\pi}{2}$ then x is equal to

(A) 0

(B) $\frac{1}{\sqrt{5}}$

(C) $\frac{2}{\sqrt{5}}$

(D) $\frac{\sqrt{3}}{2}$

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Q.5

The equation $2\cos^{-1}x = \sin^{-1}\left(2x\sqrt{1-x^2}\right)$ is valid for all values of x satisfying

(A) $-1 \le x \le 1$

(B) $0 \le x \le 1$

(C) $0 \le x \le 1/\sqrt{2}$

(D) $1/\sqrt{2} \le x \le 1$

Q.6

If $u = \cot^{-1}\sqrt{\tan\alpha} - \tan^{-1}\sqrt{\tan\alpha}$, then $\tan\left(\frac{\pi}{4} - \frac{u}{2}\right)$ is equal to

(A) $\sqrt{\tan\alpha}$

(B) $\sqrt{\cot\alpha}$

(C) $\tan\alpha$

(D) $\cot\alpha$

Q.7

Let $\tan^{-1}\left(\tan\frac{5\pi}{4}\right) = \alpha, \tan^{-1}\left(-\tan\frac{2\pi}{3}\right) = \beta$. Then

(A) $\alpha > \beta$

(B) $4\alpha - 3\beta \neq 0$

(C) $\alpha + \beta = \frac{7\pi}{12}$

(D) none of these

Q.8

If $\tan^{-1}\frac{\sqrt{1+x^2}-\sqrt{1-x^2}}{\sqrt{1+x^2}+\sqrt{1-x^2}} = \alpha$, then x^2 is equal to

(A) $\sin\alpha$

(B) $\cos 2\alpha$

(C) $\cos\alpha$

(D) $\sin 2\alpha$

Q.9

The value of $\cos^{-1}\left(-\frac{1}{2}\right) - 2\sin^{-1}\left(\frac{1}{2}\right) + 3\cos^{-1}\left(-\frac{1}{\sqrt{2}}\right) - 4\tan^{-1}(-1)$ is equal to

(A) $\frac{7\pi}{4}$

(B) $\frac{11\pi}{4}$

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(C) $\frac{\pi}{12}$

(D) $\frac{43\pi}{12}$

Q.10

The principal value of $\sin^{-1}\left(\sin\frac{2\pi}{3}\right)$ is

(A) $\frac{-2\pi}{3}$

(B) $\frac{2\pi}{3}$

(C) $\frac{4\pi}{3}$

(D) $\frac{\pi}{3}$

Q.11

If $xy+yz+zx=1$, then $\tan^{-1}x+\tan^{-1}y+\tan^{-1}z=$

(A) π

(B) $\pi/2$

(C) 1

(D) none of these

Q.12

If $2\tan^{-1}x+\sin^{-1}\frac{2x}{1+x^2}$ is independent of x, then x lies in the interval

(A) $[1,\infty[$

(B) $[-1,1]$

(C) $]-\infty,-1]$

(D) none of these

Q.13

If $k\le\sin^{-1}x+\cos^{-1}x+\tan^{-1}x\le K$, then

(A) $k=0, K=\pi$

(B) $k=\frac{\pi}{4}, K=\frac{3\pi}{4}$

(C) $k=\frac{\pi}{2}, K=\pi$

(D) both (a) and (b)

Q.14

The solution of $\tan\left(\cos^{-1}x\right)=\sin\left(\cot^{-1}\left(\frac{1}{2}\right)\right)$ is

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(A) $x = -\frac{\sqrt{5}}{4}$

(B) $x = \frac{\sqrt{5}}{4}$

(C) $x = -\frac{\sqrt{5}}{3}$

(D) $x = \frac{\sqrt{5}}{3}$

Q.15

The value of $\sin^{-1}\left(\cos\left(\sin^{-1}x\right)\right) + \cos^{-1}\left(\sin\left(\cos^{-1}x\right)\right)$ is equal to

(A) $\frac{\pi}{4}$

(B) $\frac{\pi}{2}$

(C) $\frac{3\pi}{4}$

(D) 0

Q.16

The value of $\cos\left(2\cos^{-1}0.8\right)$ is

(A) 0.48

(B) 0.96

(C) 0.6

(D) none of these

Q.17

If $\frac{1}{2}\sin^{-1}\left[\frac{3\sin 2\theta}{5+4\cos 2\theta}\right] = \tan^{-1}x$, then x is equal to

(A) $\tan 3\theta$

(B) $3\tan\theta$

(C) $\frac{1}{3}\tan\theta$

(D) $3\cot\theta$

Q.18

If $a\sin^{-1}x - b\cos^{-1}x = c$, then $a\sin^{-1}x + b\cos^{-1}x$ is equal to

(A) 0

(B) $\frac{\pi ab + c(b-a)}{a+b}$

(C) $\frac{\pi}{2}$

(D) $\frac{\pi ab + c(a-b)}{a+b}$

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Q.19

If $\cos^{-1} x - \cos^{-1} \frac{y}{2} = \alpha$, then $4x^2 - 4xy\cos\alpha + y^2$ is equal to

(A) $4\sin^2\alpha$

(B) $-4\sin^2\alpha$

(C) $2\sin 2\alpha$

(D) 4

Q.20

$\sum_{m=1}^{n} \tan^{-1} \frac{2m}{m^4+m^2+2} =$

(A) $\tan^{-1}\left(n^2+n+1\right)$

(B) $\tan^{-1}\left(n^2-n+1\right)$

(C) $\tan^{-1} \frac{n^2+n}{n^2+n+2}$

(D) none of these

Q.21

The number of solutions of $\sin^{-1} x + \sin^{-1} 2x = \frac{\pi}{3}$ is

Q.22

If $0 < \cos^{-1} x < 1$ and $1 + \sin(\cos^{-1} x) + \sin^2(\cos^{-1} x) + \sin^3(\cos^{-1} x) + \ldots\infty = 2$, then the value of $12x^2$ is ____.

Q.23

If $1 < x < \sqrt{2}$, the number of solutions of the equation $\tan^{-1}(x-1) + \tan^{-1} x + \tan^{-1}(x+1) = \tan^{-1} 3x$ is

Q.24

The value of $\sin\left\{\cos^{-1}\frac{1-x^2}{1+x^2} + \tan^{-1}\frac{1-x^2}{2x}\right\}$ is equal to λ, then $6\lambda =$

Q.25

If $\sum_{i=1}^{10} \sin^{-1} x_i = 5\pi$ then $\sum_{i=1}^{10} x_i^2$ is equal to

Q.26

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If $\cot^{-1}\frac{n}{\pi} > \frac{\pi}{6}, n \in N$, then the maximum value of n is

Q.27

The number of triplets (x, y, z) satisfying $\sin^{-1} x + \cos^{-1} y + \sin^{-1} z = 2\pi$, is

Q.28

If $\tan^{-1}\frac{1}{1+2} + \tan^{-1}\frac{1}{1+(2)(3)} + \tan^{-1}\frac{1}{1+(3)(4)} + \ldots + \tan^{-1}\frac{1}{1+n(n+1)} = \tan^{-1}\theta, \mathrm{n} = 2020, 2022\theta$ is equal to

Q.29

If $\mathrm{cosec}^{-1} x = 2\cot^{-1} 7 + \cos^{-1}(3/5)$ then the value of $117x$ is

Q.30

Least value of $\left(\sin^{-1} x\right)^3 + \left(\cos^{-1} x\right)^3$ is $\frac{-\pi^3}{n}$ then $n =$

Answers & Solutions

Q.1 Answer:

$\frac{3\pi}{4}$

Solution:

$$\text{Let } \cot^{-1}(\sqrt{2}-1) = \theta \quad \Rightarrow \quad (\sqrt{2}-1) = \cot\theta$$

$$\text{Therefore } \cos^{-1}[\cos 2\theta] = \cos^{-1}\left[\frac{1-\tan^2\theta}{1+\tan^2\theta}\right]$$

$$= \quad \cos^{-1}\left[\frac{\cot^2\theta-1}{\cot^2\theta+1}\right] = \cos^{-1}\left[\frac{2-2\sqrt{2}}{4-2\sqrt{2}}\right]$$

$$= \quad \cos^{-1}\left(-\frac{1}{\sqrt{2}}\right) = \frac{3\pi}{4}$$

Q.2 Answer:

$\left[-1, \frac{1}{\sqrt{2}}\right)$

Solution:

$$\text{Given, } \left(\cos^{-1} x\right)^2 - \left(\sin^{-1} x\right)^2 > 0$$

$$\Leftrightarrow \quad \left(\cos^{-1} x + \sin^{-1} x\right)\left(\cos^{-1} x - \sin^{-1} x\right) > 0$$

$$\Leftrightarrow \quad \frac{\pi}{2}\left(\cos^{-1} x - \sin^{-1} x\right) > 0 \quad \Leftrightarrow \quad \cos^{-1} x - \sin^{-1} x > 0$$

Q.3 Answer:

0

Solution:

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We know, $\cos^{-1}(x) + \cos^{-1}(2x)$

$= \cos^{-1}\left(x \cdot 2x - \sqrt{1-x^2}\sqrt{1-4x^2}\right)$

$= -\pi$

$x \cdot 2x - \sqrt{1-x^2}\sqrt{1-4x^2} = -1$

$2x^2 + 1 = \left(\sqrt{1-x^2}\sqrt{1-4x^2}\right)$

$4x^4 + 4x^2 + 1 = 1 + 4x^4 - 6x^2$

$10x^2 = 0$

$x = 0$

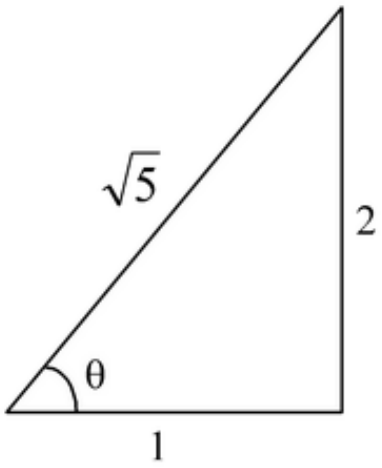
Hence there would be no real solution for the above equation.

Q.4 Answer:

$\frac{1}{\sqrt{5}}$

Solution:

Let $\cot^{-1}\frac{1}{2} = \theta \Rightarrow \cot\theta = \frac{1}{2} \Rightarrow \cos\theta = \frac{1}{\sqrt{5}} \Rightarrow \theta = \cos^{-1}\frac{1}{\sqrt{5}}$



$\therefore \sin^{-1}x + \cos^{-1}\frac{1}{\sqrt{5}} = \frac{\pi}{2} \Rightarrow \cos^{-1}\frac{1}{\sqrt{5}} = \frac{\pi}{2} - \sin^{-1}x = \cos^{-1}x \Rightarrow x = \frac{1}{\sqrt{5}}$

Q.5 Answer:

$1/\sqrt{2} \le x \le 1$

Solution:

If we denote $\cos^{-1}x$ **by** y**, then**

since $0 \le \cos^{-1}x \le \pi \quad \Rightarrow \quad 0 \le 2y \le 2\pi$ **...(i)**

Also since $-\frac{\pi}{2} \le \sin^{-1}\left(2x\sqrt{1-x^2}\right) \le \frac{\pi}{2}$

$\Rightarrow -\frac{\pi}{2} \le 2y \le \frac{\pi}{2}$ **...(ii)**

From (i) and (ii) we find $0 \le 2y \le \frac{\pi}{2}$

$\Rightarrow \quad 0 \le y \le \frac{\pi}{4}$

$\Rightarrow 0 \le \cos^{-1}x \le \frac{\pi}{4}$

which holds if $1/\sqrt{2} \le x \le 1$.

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Q.6 Answer:

$\sqrt{\tan\alpha}$

Solution:

Let $\sqrt{\tan\alpha}=\tan x$, then $u=\cot^{-1}(\tan x)-\tan^{-1}(\tan x)$
$=\frac{\pi}{2}-x-x=\frac{\pi}{2}-2x$
$\Rightarrow \quad = \quad 2x=\frac{\pi}{2}-u \quad \Rightarrow \quad x=\frac{\pi}{4}-\frac{u}{2}$
$\Rightarrow \tan x=\tan\left(\frac{\pi}{4}-\frac{u}{2}\right)$
$\Rightarrow \quad \sqrt{\tan\alpha}=\tan\left(\frac{\pi}{4}-\frac{u}{2}\right)$

Q.7 Answer:

$\alpha+\beta=\frac{7\pi}{12}$

Solution:

$\tan^{-1}\left\{\tan\left(\pi+\frac{\pi}{4}\right)\right\}=\tan^{-1}\tan\frac{\pi}{4}=\frac{\pi}{4}=\alpha$
$\tan^{-1}\left\{-\tan\left(\pi-\frac{\pi}{3}\right)\right\}=\tan^{-1}\tan\frac{\pi}{3}=\frac{\pi}{3}=\beta$

Q.8 Answer:

$\sin 2\alpha$

Solution:

From the given relation we have $\frac{\sqrt{1+x^2}-\sqrt{1-x^2}}{\sqrt{1+x^2}+\sqrt{1-x^2}}=\tan\alpha$
$\Rightarrow \frac{2\sqrt{1+x^2}}{2\sqrt{1-x^2}}=\frac{1+\tan\alpha}{1-\tan\alpha}$ (by componendo and dividendo)
$\Rightarrow \quad \frac{1+x^2}{1-x^2}=\left(\frac{\sin\alpha+\cos\alpha}{\cos\alpha-\sin\alpha}\right)^2=\frac{1+\sin 2\alpha}{1-\sin 2\alpha}\Rightarrow x^2=\sin 2\alpha \quad$ [Using (1)]

Q.9 Answer:

$\frac{43\pi}{12}$

Solution:

We have $\cos^{-1}\left(-\frac{1}{2}\right)-2\sin^{-1}\frac{1}{2}+3\cos^{-1}\left(-\frac{1}{\sqrt{2}}\right)-4\tan^{-1}(-1)$
$=\frac{2\pi}{3}-2\times\frac{\pi}{6}+3\times\frac{3\pi}{4}-4\left(\frac{-\pi}{4}\right)=\frac{43\pi}{12}$

Q.10 Answer:

$\frac{\pi}{3}$

Solution:

$\sin^{-1}\left(\sin\frac{2\pi}{3}\right)=\sin^{-1}\sin\left(\pi-\frac{\pi}{3}\right)$
$=\sin^{-1}\sin\frac{\pi}{3}=\frac{\pi}{3}$

Q.11 Answer:

$\pi/2$

Solution:

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$\tan^{-1} x + \tan^{-1} y + \tan^{-1} z$
$= \tan^{-1}\left(\frac{x+y}{1-xy}\right) + \tan^{-1} z$
$= \tan^{-1}\left(\frac{x+y+x-xyz}{1-xy-yz-zx}\right)$
$= \tan^{-1} \infty \quad [\text{Since } xy + yz + zx = 1]$
$= \frac{\pi}{2}$

Q.12 Answer:

$[1, \infty[$

Solution:

$\sin^{-1} \frac{2x}{1+x^2} = 2\tan^{-1} x$, **if** $|x| \leq 1$ **and** $\pi - 2\tan^{-1} x$, **if** $|x| \geq 1$
$\therefore$ **When** $|x| \geq 1, \quad 2\tan^{-1} x + \sin^{-1} \frac{2x}{1+x^2} = \pi$ **...(i)**
Which is independent of x.
But if $x < 0$, **then L.H.S. of (i) is negative which is not acceptable.**
$\therefore \quad x > 0$ **and** $|x| \geq 1 \Rightarrow x \geq 1$

Q.13 Answer:

$k = \frac{\pi}{4}, K = \frac{3\pi}{4}$

Solution:

We have, $\sin^{-1} x + \cos^{-1} x + \tan^{-1} x = \frac{\pi}{2} + \tan^{-1} x$
Domain of above function is $[-1, 1]$
Since, $-1 \leq x \leq 1 \Rightarrow -\frac{\pi}{4} \leq \tan^{-1} x \leq \frac{\pi}{4} \Rightarrow \quad \frac{\pi}{2} - \frac{\pi}{4} \leq \frac{\pi}{2} + \tan^{-1} x \leq \frac{\pi}{2} + \frac{\pi}{4}$
So, $k = \frac{\pi}{4}, K = \frac{3\pi}{4}$

Q.14 Answer:

$x = \frac{\sqrt{5}}{3}$

Solution:

Given $\tan\left(\cos^{-1} x\right) = \sin\left(\cot^{-1}\left(\frac{1}{2}\right)\right)$
Let $\cos^{-1} x = \theta$ **and** $\cot^{-1}(1/2) = \varphi$
$\Rightarrow \quad \tan\theta = \sin\varphi \quad \Rightarrow \frac{\sqrt{1-x^2}}{x} = \frac{2}{\sqrt{5}} \Rightarrow 1 - x^2 = \frac{4x^2}{5} \Rightarrow x^2 = \frac{5}{9} \Rightarrow x = \pm\frac{\sqrt{5}}{3}$
but $x = -\frac{\sqrt{5}}{3}$ **does not satisfy the above equation, so** $x = \frac{\sqrt{5}}{3}$

Q.15 Answer:

$\frac{\pi}{2}$

Solution:

Given $\sin^{-1}\left(\cos\left(\sin^{-1} x\right)\right) + \cos^{-1}\left(\sin\left(\cos^{-1} x\right)\right)$
$= \sin^{-1}\left[\cos\left(\frac{\pi}{2} - \cos^{-1} x\right)\right] + \cos^{-1}\left[\sin\left(\frac{\pi}{2} - \sin^{-1} x\right)\right]$

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$= \sin^{-1}\left[\sin\left(\cos^{-1} x\right)\right] + \cos^{-1}\left[\cos\left(\sin^{-1} x\right)\right] = \cos^{-1} x + \sin^{-1} x = \frac{\pi}{2}$

Q.16 Answer:

none of these

Solution:

Since, $2\cos^{-1} x = \cos^{-1}\left(2x^2 - 1\right)$**. Therefore,**

$$2\cos^{-1} 0.8 = \cos^{-1}(2 \times 0.64 - 1) = \cos^{-1}(0.28)$$

$$\Rightarrow \quad \cos\left(2\cos^{-1} 0.8\right) = \cos\left(\cos^{-1} 0.28\right) = 0.28$$

Q.17 Answer:

$\frac{1}{3}\tan\theta$

Solution:

Given $\left[\frac{3\sin 2\theta}{5+4\cos 2\theta}\right] = \frac{6\tan\theta}{9+\tan^2\theta} = \frac{2\tan\phi}{1+\tan^2\phi} = \sin 2\phi$

where $\tan\theta = 3\tan\phi$

$\frac{1}{2}\sin^{-1}\left[\frac{3\sin 2\theta}{5+4\cos 2\theta}\right] = \phi = \tan^{-1}\left[\left(\frac{1}{3}\right)\tan\theta\right]$

$\Rightarrow \quad x = \frac{1}{3}\tan\theta$

Q.18 Answer:

$\frac{\pi ab + c(a-b)}{a+b}$

Solution:

Given $b\sin^{-1} x + b\cos^{-1} x = \frac{b\pi}{2}$

and $a\sin^{-1} x - b\cos^{-1} x = c$

$\Rightarrow \quad (a+b)\sin^{-1} x = \frac{b\pi}{2} + c \quad \Rightarrow \quad \sin^{-1} x = \frac{\frac{b\pi}{2}+c}{a+b}$

Similarly, $\cos^{-1} x = \frac{\frac{a\pi}{2}-c}{a+b}$

So that $a\sin^{-1} x + b\cos^{-1} x = \frac{\pi ab + c(a-b)}{a+b}$

Q.19 Answer:

$4\sin^2\alpha$

Solution:

$\cos\alpha = x(y/2) + \sqrt{(1-x^2)}\sqrt{1-(y^2/4)}$

$\Rightarrow \quad (2\cos\alpha - xy)^2 = \left(1-x^2\right)\left(4-y^2\right)$

$\Rightarrow \quad 4\cos^2\alpha - 4xy\cos\alpha = 4 - 4x^2 - y^2$

$\Rightarrow \quad 4x^2 - 4xy\cos\alpha + y^2 = 4\left(1-\cos^2\alpha\right) = 4\sin^2\alpha.$

Q.20 Answer:

$\tan^{-1}\frac{n^2+n}{n^2+n+2}$

Solution:

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$\sum_{m=1}^{n} \tan^{-1} \frac{2m}{m^4+m^2+2}$
$= \sum_{m=1}^{n} \tan^{-1} \frac{1+m+m^2-\left(1-m+m^2\right)}{1+(1+m+m^2)(1-m+m^2)}$
$= \sum_{m=1}^{n} \left[\tan^{-1}\left(1+m+m^2\right) - \tan^{-1}\left(1-m+m^2\right)\right]$
$= \sum_{m=1}^{n} \left[\tan^{-1}\left(1+m+m^2\right) - \tan^{-1}\left(1+(m-1)+(m-1)^2\right)\right]$
$= \tan^{-1}\left(1+n+n^2\right) - \tan^{-1} 1$
$= \tan^{-1} \frac{n+n^2}{1+1+n+n^2} = \tan^{-1} \frac{n+n^2}{2+n+n^2}$

Q.21 Answer:

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Solution:

$\sin^{-1} x + \sin^{-1} 2x = \frac{\pi}{3} \Rightarrow \quad \cos^{-1} x + \cos^{-1} 2x = \frac{2\pi}{3}$
$\Rightarrow \cos^{-1}\left(2x^2 - \sqrt{1-x^2}\sqrt{1-4x^2}\right) = \frac{2\pi}{3} \Rightarrow x = \pm\frac{1}{2}\sqrt{\frac{3}{7}}$ **(but x can't be negative)**

Q.22 Answer:

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Solution:

$\mathrm{S} = \frac{\mathrm{a}}{1-\mathrm{r}}$
$= \frac{1}{1-\sin(\cos^{-1}(x))} \ldots$ **(i)**
Let $\cos^{-1}(\mathrm{x}) = \mathrm{y}$
$\therefore \cos y = x$
$\sin y = \pm\sqrt{1-x^2}$
$\mathrm{y} = \sin^{-1}\left(\pm\sqrt{1-\mathrm{x}^2}\right)$
$= \pm \sin^{-1}\left(\sqrt{1-x^2}\right)$
Substituting in the equation, gives us
$= \frac{1}{1\pm\sqrt{1-x^2}}$
$= 2$
$2 \pm 2\sqrt{1-\mathrm{x}^2} = 1$
$2 - 1 = \pm 2\sqrt{1-\mathrm{x}^2}$
$\frac{1}{2} = \pm\sqrt{1-\mathrm{x}^2}$
Squaring both sides, we get
$\frac{1}{4} = 1 - \mathrm{x}^2$
$x^2 = \frac{3}{4}$
Hence, $12\mathrm{x}^2 = 12 \times \frac{3}{4} = 9.$

Q.23 Answer:

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Solution:

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The given equation can be written as

$$\tan^{-1}(x-1)+\tan^{-1}(x+1)=\tan^{-1}3x-\tan^{-1}x$$
$$\Rightarrow \quad \tan^{-1}\frac{x-1+x+1}{1-(x-1)(x+1)}=\tan^{-1}\frac{3x-x}{1+3x^2}\Rightarrow\frac{2x}{2-x^2}=\frac{2x}{1+3x^2}\Rightarrow x+3x^3=2x-x^3$$
$$\Rightarrow \quad 4x^3-x=0\Rightarrow x\left(4x^2-1\right)=0\Rightarrow x=0,x=\pm\frac{1}{2}$$

none of which satisfies $1<x<\sqrt{2}$

Q.24 Answer:

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Solution:

Put $x=\tan\theta$

$$\therefore\cos^{-1}\frac{1-x^2}{1+x^2}=\cos^{-1}(\cos 2\theta)=2\theta$$
$$\tan^{-1}\frac{1-x^2}{2x}=\tan^{-1}(\cot 2\theta)=\tan^{-1}\tan\left(\frac{\pi}{2}-2\theta\right)=\frac{\pi}{2}-2\theta$$
$$\therefore\sin\left[\cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)+\tan^{-1}\left(\frac{1-x^2}{2x}\right)\right]=\sin\left(2\theta+\frac{\pi}{2}-2\theta\right)=\sin\frac{\pi}{2}=1$$

Q.25 Answer:

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Solution:

$$\sum_{i=1}^{10}\sin^{-1}x_i=5\pi=10\cdot\frac{\pi}{2}$$
$$\Rightarrow \quad \sin^{-1}x_i=\frac{\pi}{2},\forall i \quad \Rightarrow \quad \sum_{i=1}^{10}x_i^2=10$$

Q.26 Answer:

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Solution:

$$\cot^{-1}\frac{n}{\pi}>\frac{\pi}{6} \quad \Rightarrow \quad \cot\left(\cot^{-1}\frac{n}{\pi}\right)<\cot\frac{\pi}{6}$$
$$\Rightarrow \quad \frac{n}{\pi}<\sqrt{3} \quad \Rightarrow \quad n<\sqrt{3}\pi=5.5 \text{ (nearly).}$$

So, the maximum value of n **is 5.**

Q.27 Answer:

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Solution:

The given equation is $\sin^{-1}x+\cos^{-1}y+\sin^{-1}z=2\pi$

Given equation exist only, when $\sin^{-1}x=\frac{\pi}{2},\cos^{-1}y=\pi$ **and** $\sin^{-1}z=\frac{\pi}{2}$

$\Rightarrow x=1,y=-1$ **and** $z=1 \quad \Rightarrow (x,y,z)\equiv(1,-1,1)$

Q.28 Answer:

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Solution:

$\tan^{-1}\frac{1}{1+n(n+1)}=\tan^{-1}\frac{n+1-n}{1+n(n+1)}=\tan^{-1}(n+1)-\tan^{-1}(n)$

so that L.H.S. of the given equation is

$=\tan^{-1}2-\tan^{-1}1+\tan^{-1}3-\tan^{-1}2+\ldots.+\tan^{-1}(n+1)-\tan^{-1}n$

$=\tan^{-1}(n+1)-\tan^{-1}1=\tan^{-1}\frac{n+1-1}{1+(n+1)}=\tan^{-1}\frac{n}{n+2}$

so that $\tan^{-1}\frac{n}{n+2}=\tan^{-1}\theta\Rightarrow\theta=\frac{n}{n+2}$

Q.29 Answer:

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Solution:

$2\cot^{-1}7+\cos^{-1}\frac{3}{5}=\cot^{-1}\frac{7^2-1}{2\times7}+\cot^{-1}\frac{3}{4}$

$=\cot^{-1}\frac{24}{7}+\cot^{-1}\frac{3}{4}$

$=\cot^{-1}\frac{\frac{24}{7}\times\frac{3}{4}-1}{\frac{24}{7}+\frac{3}{4}}=\cot^{-1}\frac{44}{117}=\operatorname{cosec}^{-1}\frac{125}{117}$

Q.30 Answer:

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Solution:

Given expression $y=\left(\sin^{-1}x+\cos^{-1}x\right)^3$

$-3\sin^{-1}x\cos^{-1}x\left(\sin^{-1}x+\cos^{-1}x\right)=\left(\frac{\pi}{2}\right)^3-3ab\cdot\frac{\pi}{2}$

where $a=\sin^{-1}x, b=\cos^{-1}x$

$=\frac{\pi^3}{8}-\frac{3}{8}\pi\left[(a+b)^2-(a-b)^2\right]=\frac{\pi^3}{8}-\frac{3}{8}\pi\left[\frac{\pi^2}{4}-(a-b)^2\right]=\frac{3}{8}\pi(a-b)^2+\frac{\pi^3}{32}$

For y **to be minimum** $(a-b)^2=0$ **or** $a=b$

or $\sin^{-1}x=\cos^{-1}x=\frac{\pi}{2}-\sin^{-1}x$ **or** $\sin^{-1}x=\frac{\pi}{4}$ **or** $x=\sin\frac{\pi}{4}=\frac{1}{\sqrt{2}}$

Minimum value of $y=\frac{\pi^3}{32}$

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